

Simulation Exclusion Module (SEXM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Purpose & Scope Standard (SPSS)

Operational Layer: Simulation Purpose & Scope Governance Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Parent Standard Module

Governed Space: Simulation Exclusions

Version: 1.0

Status: Canonical · Open Module

Effective Date: 16 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Simulation Exclusions

Canonical Definition

Simulation Exclusion Module (SEXM) defines the governance conditions under which simulation-relevant objects, functions, behaviors, interactions, conditions, environments, populations, scenarios, temporal periods, failure modes, capabilities, variables, dependencies, and other material elements are explicitly identified as outside the governed Simulation Scope and preserved as visible limitations of the simulation inquiry.

SEXM establishes the formal negative boundary of a simulation: not merely what the simulation covers, but what it does not cover and therefore cannot legitimately be assumed to have simulated.

Operational Role

SEXM serves as the third internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The preceding modules establish:

SPM — Why is the simulation being performed?

SSCM — What exactly must the simulation cover?

SEXM asks:

What remains outside that scope, why is it excluded, and what consequences does the exclusion create for simulation interpretation?

The governed progression is:

Simulation Purpose → Simulation Scope → Candidate Exclusions → Exclusion Classification → Exclusion Justification → Materiality Assessment → Exclusion Consequence → Governed Simulation Exclusions

SEXM prevents absence from becoming invisible.

Module Operational Space

SEXM governs: Simulation Exclusion identification, explicit exclusions, implicit-exclusion detection, object exclusions, component exclusions, functional exclusions, behavioral exclusions, interaction exclusions, environmental exclusions, operational-condition exclusions, spatial exclusions, temporal exclusions, population exclusions, capability exclusions, autonomy exclusions, scenario exclusions, event exclusions, failure-mode exclusions, variable exclusions, dependency exclusions, boundary-condition exclusions, data-related scope exclusions, excluded uncertainty domains, exclusion rationale, exclusion classification, exclusion materiality, exclusion dependencies, exclusion consequences, exclusion conflicts, exclusion change, exclusion traceability, and Simulation Exclusion status.

Module Function

A simulation can appear broader than it actually is when omitted elements remain undocumented.

SEXM exists to prevent:

Not Simulated → Not Mentioned → Assumed Covered

from becoming an acceptable simulation-governance pathway.

Its function is to transform material absence into governed information.

The required relationship becomes:

Excluded Element → Reason for Exclusion → Materiality → Affected Scope → Interpretive Consequence

An exclusion does not automatically make a simulation inadequate.

An undisclosed material exclusion may.

Simulation Exclusion

A Simulation Exclusion is an identifiable simulation-relevant element intentionally or necessarily maintained outside the governed Simulation Scope.

An exclusion may concern: part of the Simulation Object, a system function, a behavior, an interaction, an environment, an operating condition, a population, a time period, a scenario, a failure mode, a dependency, or another materially relevant dimension.

Positive and Negative Simulation Boundaries

SSCM establishes the positive boundary:

What is inside the simulation inquiry?

SEXM establishes the negative boundary:

What is outside the simulation inquiry?

Together they create a more complete scope structure:

Included Space | Governed Boundary | Excluded Space

A simulation boundary is methodologically stronger when both sides are visible.

Explicit Exclusion Principle

SIMULOS® establishes:

A material element outside Simulation Scope should remain explicitly identifiable where its absence could affect interpretation of the simulation or its outputs.

Silence is not a sufficient exclusion mechanism.

Exclusion Does Not Mean Irrelevance

An excluded element may remain operationally important.

For example:

Snow conditions

may be excluded from an autonomous-vehicle simulation because the current simulation campaign addresses dry and rainy environments.

This does not establish:

Snow is irrelevant.

It establishes:

Snow is outside the current governed Simulation Scope.

Exclusion Classification

Simulation Exclusions may be classified according to why they exist.

Possible classes include: Purpose-Based Exclusion

The element is not required for the governed Simulation Purpose.

Scope-Based Exclusion

The element falls outside the explicitly approved Simulation Scope.

Capability-Based Exclusion

The available simulation environment cannot adequately represent the element.

Evidence-Based Exclusion

Required inputs or supporting information are insufficient.

Methodological Exclusion

The simulation methodology is not suitable for representing the element.

Resource-Based Exclusion

The element is omitted because of time, computational, financial, infrastructure, or other resource constraints.

Dependency-Based Exclusion

The element cannot be simulated because a required dependency is unavailable or unresolved.

Uncertainty-Based Exclusion

The element cannot be represented with sufficient confidence for the intended inquiry.

Regulatory or Governance Exclusion

Applicable governance conditions prevent or constrain its inclusion.

Deferred Exclusion

The element is intentionally reserved for a later simulation cycle.

The classification should make clear why the exclusion exists.

Intentional Exclusion

An Intentional Exclusion is deliberately established before or during simulation planning.

Intentional exclusions should remain traceable to their rationale.

A deliberate exclusion may be completely legitimate where it is compatible with the Simulation Purpose.

Necessary Exclusion

A Necessary Exclusion exists where an element cannot reasonably be included under the current simulation conditions.

For example: unavailable data, unavailable model capability, unresolved physics, unavailable environmental representation, inaccessible dependencies, or insufficient computational capability.

Necessary exclusions remain limitations even when unavoidable.

Deferred Exclusion

A material element may be excluded from one simulation cycle because it will be addressed separately.

For example:

Nominal Operation Simulation

may precede:

Failure & Recovery Simulation

The first simulation must not be represented as having covered the deferred failure conditions.

Object Exclusion

Parts of the Simulation Object may remain outside scope.

For example:

A spacecraft simulation may include:

Guidance + Navigation + Propulsion

while excluding:

Crew Life-Support Dynamics

SEXM preserves that distinction.

Functional Exclusion

A system may possess functions not examined by the simulation.

These functions should remain identifiable where their omission affects interpretation.

Behavioral Exclusion

Behavioral modes may be excluded.

Examples include: degraded behavior, emergency behavior, adversarial behavior, adaptive behavior, rare behavior, or human intervention.

Simulation of nominal behavior does not imply simulation of excluded behaviors.

Interaction Exclusion

A simulation may represent multiple actors while excluding certain interactions among them.

This distinction matters because:

Objects Represented ≠ All Interactions Represented

For example, two autonomous agents may both exist in a simulation while agent-to-agent negotiation remains excluded.

Environmental Exclusion

Environmental conditions may materially affect simulation outcomes.

Possible exclusions include: extreme temperature, rain, snow, radiation, low visibility, high wind, electromagnetic interference, network degradation, terrain classes, or other operational environments.

Environmental exclusions should remain connected to output applicability.

Operational Condition Exclusion

A simulation may cover:

Normal Operation

while excluding:

Emergency Operation

or:

Maximum Load Conditions

Such exclusions must remain visible wherever simulation outputs could otherwise be interpreted more broadly.

Spatial Exclusion

Simulation applicability may exclude specific geographic, physical, or logical regions.

A simulation performed for one operating region must not silently imply coverage of excluded regions.

Temporal Exclusion

A simulation may exclude: startup periods, long-term degradation, seasonal variation, future system states, specific mission phases, or extended operating durations.

A simulation covering short-term behavior cannot establish long-term behavior where long-term dynamics remain excluded.

Population Exclusion

Where humans, users, agents, organizations, or other populations are relevant, specific groups may remain outside scope.

For example:

Adult Population Included

Pediatric Population Excluded

The exclusion must remain visible wherever outputs could otherwise be generalized.

Capability Exclusion

A simulation of an AI or autonomous system may intentionally exclude capabilities.

For example:

Navigation Included

Autonomous Tool Execution Excluded

Simulation results must remain bounded accordingly.

Autonomy Exclusion

A simulation may examine a system under:

Human Supervision

while excluding:

Fully Autonomous Operation

The system's broader technical capability does not remove this exclusion.

Scenario Exclusion

Not every possible scenario must be simulated.

SEXM governs which materially relevant scenario classes remain outside the simulation campaign.

Examples may include: rare events, edge cases, adversarial scenarios, extreme-load scenarios, cascading failures, or multi-system emergencies.

Failure-Mode Exclusion

Failure modes may be excluded from a simulation.

Where safety, resilience, recovery, or reliability conclusions are intended, such exclusions may become highly material.

SEXM does not determine whether a failure mode must be simulated.

It ensures that exclusion remains visible and interpretable.

Dependency Exclusion

A simulation may exclude a dependency or simplify its role.

Examples include: external infrastructure, third-party systems, communications networks, human operators, supply dependencies, external AI services, or environmental dependencies.

The excluded dependency may still materially influence real-world behavior.

Variable Exclusion

Some variables may be held constant, omitted, or intentionally removed from the simulation inquiry.

Where the excluded variable could materially affect outcomes, the exclusion should remain explicit.

Excluded Uncertainty

A simulation may deliberately avoid or defer uncertainty domains that cannot yet be sufficiently represented.

SEXM requires such uncertainty exclusions to remain visible rather than being converted into artificial certainty.

Implicit Exclusion

An Implicit Exclusion exists where an element is outside actual simulation coverage but has not been formally declared outside scope.

Implicit exclusions are methodologically dangerous because they can create false assumptions of coverage.

Implicit-Exclusion Detection

SEXM should support comparison between:

Simulation Purpose

Governed Simulation Scope

Planned Representation

Scenario Coverage

and:

Actual Simulation Coverage

to identify material elements that have disappeared without explicit exclusion.

Unknown Exclusion

Some relevant conditions may not be known when Simulation Scope is established.

Where later analysis identifies a previously unknown omitted condition, it should become a governed exclusion, scope gap, or scope-change question rather than remaining invisible.

Exclusion Justification

Material exclusions should have a documented basis.

Possible justifications include: not relevant to current purpose, outside approved scope, separately simulated, technically unavailable, insufficient evidence, insufficient model capability, unacceptable uncertainty, deferred to future simulation, or resource limitation.

A justification explains the exclusion.

It does not automatically make the exclusion acceptable for every intended interpretation.

Exclusion Materiality

Not all exclusions have equal significance.

An exclusion is material where its absence could reasonably affect: simulation design, output interpretation, comparison, applicability, decision relevance, safety relevance, reliance, or downstream validation.

Materiality should be assessed relative to the governed Simulation Purpose and Scope.

Material Exclusion

A Material Simulation Exclusion is an exclusion capable of changing the meaning, applicability, or interpretation of simulation outputs.

Material exclusions should propagate downstream.

Non-Material Exclusion

An exclusion may be non-material where its absence cannot reasonably affect the governed inquiry.

Non-materiality should not be inferred merely because an excluded element produced no observed problem.

Exclusion Consequence

Each material exclusion should identify the consequence it creates.

Possible consequences include:

No Material Consequence

Interpretation Limitation

Applicability Limitation

Scenario Limitation

Coverage Limitation

Representation Limitation

Output Qualification Required

Additional Simulation Required

or:

Consequence Undetermined Exclusion

Propagation

Material exclusions should remain available to downstream SIMULOS® governance.

Conceptually:

Exclusion → Reality Representation → Assumptions → Model & Environment →
Scenario Design → Execution → Output Governance → Reality Correlation

This prevents an exclusion declared early in the lifecycle from disappearing before outputs are interpreted.

Exclusion Dependency

One exclusion may create additional exclusions.

For example:

External Communication Network Excluded

may also require exclusion of:

Network-Failure Recovery Behavior

where that behavior depends upon representation of the network.

SEXM should preserve such dependency chains.

Compound Exclusions

Several individually limited exclusions may collectively become material.

For example:

Night Excluded

Rain Excluded

High-Speed Operation Excluded

Emergency Braking Excluded

may collectively make a broad claim of:

Urban Autonomous Driving Simulation

methodologically misleading.

SEXM therefore permits cumulative exclusion assessment. Exclusion–Purpose Conflict

An Exclusion–Purpose Conflict exists where an excluded element is materially required to answer the governed Simulation Purpose.

Example:

Simulation Purpose: Evaluate emergency landing safety.

Exclusion: Hydraulic failure conditions.

If hydraulic failure is materially relevant to emergency landing safety, the exclusion creates a direct purpose conflict. Exclusion–Scope Conflict

An exclusion conflicts with Simulation Scope where an element is simultaneously represented as:

inside governed scope

and:

excluded from actual simulation coverage.

Such a conflict must be resolved or preserved explicitly as a scope gap.

Exclusion Cannot Be Hidden by Abstraction

Simplifying a material element through abstraction is not necessarily equivalent to excluding it.

However, where abstraction removes the material behavior the inquiry depends upon, the architecture should prevent the simplification from being used to disguise functional exclusion.

Reality Representation governance addresses the quality of representation.

SEXM governs whether the relevant element is actually inside or outside the simulation inquiry.

Exclusion Cannot Become Assumption by Default

An excluded element and an assumed condition are not the same governance concept.

For example:

Wind Excluded

is different from:

Wind Assumed Constant at 5 m/s

The first removes wind dynamics from scope.

The second represents wind through an assumption.

SEXM preserves this distinction while later Assumption governance governs the assumption itself.

Exclusion Cannot Become Validation

The fact that an excluded condition was not simulated cannot support a positive conclusion about that condition.

SIMULOS® establishes:

Not Simulated ≠ Successfully Simulated

and:

Excluded ≠ Validated No Silent Exclusion Rule

A material simulation element must not disappear from the governed inquiry without becoming either explicitly excluded, represented as a scope gap, or returned for scope reassessment.

No Exclusion Concealment Rule

An exclusion must not be hidden because disclosure would weaken the perceived completeness of a simulation.

No Convenience Relevance Rule

An element must not be declared irrelevant solely because: it is difficult to model, data are unavailable, computation is expensive, the current tool cannot represent it, or its inclusion produces unfavorable outcomes.

These conditions may justify exclusion.

They do not establish irrelevance.

No Exclusion-to-Coverage Conversion Rule

A collection of successfully simulated conditions cannot convert excluded conditions into covered conditions.

Covered Scope + Excluded Scope ≠ Complete Scope

unless the exclusions are genuinely outside the required inquiry.

No Retrospective Exclusion Rule

A condition must not be retrospectively declared excluded merely because its simulation produced undesirable results.

Where a condition was within the governed Simulation Scope during execution, removing it after results are known constitutes a material scope change and must be governed accordingly.

No Exclusion Generalization Rule

Results obtained inside included conditions must not automatically be generalized to excluded conditions.

Exclusion Change

Simulation Exclusions may change.

An element may move:

Excluded → Included

or:

Included → Excluded

Such changes can alter downstream simulation requirements.

Exclusion Removal

When an excluded element becomes included, its removal from the exclusion set should trigger assessment of whether new: representation, assumptions, model capability, inputs, scenarios, execution, or output interpretation

are required.

New Exclusion

A new exclusion introduced after simulation planning may materially narrow the inquiry.

The resulting scope and downstream applicability should be reassessed.

Exclusion Status

Possible statuses may include:

Established Exclusion

Conditionally Excluded

Deferred

Temporarily Excluded

Exclusion Under Review

Exclusion Conflicted

or:

Exclusion Undetermined Exclusion Register

A governed simulation may maintain a Simulation Exclusion Register.

The register may identify: Exclusion ID, excluded element, exclusion class, rationale, affected scope, materiality, dependencies, downstream consequences, status, effective point, and change history.

Exclusion Traceability

Material exclusions should remain traceable through simulation outputs.

A reviewer examining an output should be capable of determining whether a relevant condition was:

Included

Excluded

Not Represented

or:

Undetermined

where that distinction materially affects interpretation.

Failure Conditions

SEXM governance failure may exist where: material exclusions are undocumented, excluded conditions are represented as simulated, absence is treated as successful coverage, an exclusion conflicts with the Simulation Purpose without disclosure, exclusions conflict with declared Simulation Scope, technical inability is presented as irrelevance, resource constraints are hidden, material excluded dependencies remain invisible, cumulative exclusions materially narrow the simulation without disclosure, excluded populations are generalized into represented populations, excluded operating conditions are treated as covered, retrospective exclusions remove unfavorable results, excluded uncertainty is converted into apparent certainty, exclusions fail to propagate to downstream interpretation, or simulation claims extend into explicitly excluded space.

Minimum Implementation Framework

1. Identify Excluded Simulation Elements

Determine which materially relevant objects, functions, behaviors, conditions, interactions, environments, populations, scenarios, dependencies, and other elements remain outside the governed Simulation Scope. 2. Classify and Justify Exclusions

Establish why each material exclusion exists and distinguish purpose-based, capability-based, methodological, resource-based, deferred, dependency-based, and other applicable exclusion classes. 3. Assess Materiality and Consequence

Determine whether each exclusion affects simulation coverage, representation, interpretation, applicability, downstream validation, or required additional

simulation. 4. Govern Exclusion Dependencies and Change

Preserve cumulative exclusions, dependency effects, inclusion/exclusion transitions, conflicts, and material changes. 5. Preserve Exclusion Traceability

Maintain a Simulation Exclusion Register and propagate material exclusions into downstream simulation design, execution, outputs, and interpretation.

Governance Outputs

SEXM may produce: Simulation Exclusion Definition, Simulation Exclusion Register, Simulation Exclusion Record, Exclusion Classification Record, Exclusion Rationale Record, Exclusion Materiality Assessment, Exclusion Consequence Record, Purpose-Based Exclusion Record, Scope-Based Exclusion Record, Capability-Based Exclusion Record, Methodological Exclusion Record, Resource-Based Exclusion Record, Evidence-Based Exclusion Record, Dependency-Based Exclusion Record, Uncertainty-Based Exclusion Record, Deferred Exclusion Record, Object Exclusion Record, Functional Exclusion Record, Behavioral Exclusion Record, Interaction Exclusion Record, Environmental Exclusion Record, Operational Condition Exclusion Record, Spatial Exclusion Record, Temporal Exclusion Record, Population Exclusion Record, Capability Exclusion Record, Autonomy Exclusion Record, Scenario Exclusion Record, Failure-Mode Exclusion Record, Dependency Exclusion Record, Variable Exclusion Record, Implicit Exclusion Record, Unknown Exclusion Record, Compound Exclusion Assessment, Exclusion–Purpose Conflict Record, Exclusion–Scope Conflict Record, Exclusion Change Record, Exclusion Removal Record, New Exclusion Record, Simulation Exclusion Status, and Simulation Exclusion Traceability Map.

Use Case 1 — Autonomous Robot Safety Simulation

Scenario

A mobile industrial robot is simulated to examine collision avoidance within a warehouse.

The simulation covers:

normal warehouse traffic,

human workers,

standard lighting,

and:

defined operating speeds.

It does not represent emergency power loss, smoke-obscured sensors, or operation outside the warehouse. Application

SEXM formally establishes these as Simulation Exclusions.

Emergency power loss is classified as a Deferred Exclusion scheduled for a

separate safety simulation.

Smoke-obscured perception is classified as a Capability-Based Exclusion because the current environment cannot adequately represent the sensor degradation.

Outdoor operation is a Purpose-Based Exclusion because it falls outside the governed Simulation Purpose. Result

The simulation may support interpretation within its governed warehouse conditions.

It cannot legitimately be presented as comprehensive simulation of the robot across emergency, degraded-visibility, and outdoor environments. Use Case 2 — Spacecraft Mission Simulation

Scenario

A spacecraft simulation examines orbital insertion and autonomous navigation.

Solar radiation effects are simplified, crew behavior is excluded, and one rare propulsion-failure mode is not represented because the required physical model is unavailable. Application

SEXM distinguishes the three conditions.

Simplified solar radiation remains inside the simulation and is governed through Reality Representation and Assumption governance.

Crew behavior is an explicit Scope Exclusion.

The propulsion failure becomes a Capability-Based Material Exclusion because the simulation environment cannot currently represent it. Result

The architecture prevents all three conditions from being treated identically.

The simulation record can distinguish:

represented through simplification,

explicitly excluded,

and:

materially excluded because simulation capability is insufficient.

This distinction remains available when simulation outputs are later interpreted.

Architectural Position

Simulation Exclusion Module (SEXM) is the third internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The internal progression is:

Simulation Purpose → Simulation Scope → Simulation Exclusions → Simulation Applicability Context → Simulation Depth & Precision

SPM establishes why the simulation exists.

SSCM establishes what the simulation must cover.

SEXM establishes what remains outside that governed inquiry and what consequences follow from that absence.

The subsequent modules determine where the resulting simulation is contextually applicable and how deeply and precisely the simulation must represent the intended inquiry.

SEXM does not govern general integrity under INTEGROS®, Validation Scope under VALIDOS®, actual Operational Reality under ORA™, or the fidelity of Reality Representation under RRS.

It governs specifically:

the explicit negative boundary of the simulation inquiry.

Foundational Principle

A simulation is governed not only by what it includes, but by whether everything materially excluded from the inquiry remains visible wherever that absence can affect the meaning of its results.

Canonical Closing Statement

Simulation Exclusion Module (SEXM) establishes the negative boundary of governed simulation by identifying, classifying, justifying, assessing, and preserving the objects, functions, behaviors, interactions, environments, populations, scenarios, failure modes, dependencies, uncertainties, and other material elements that remain outside Simulation Scope. Through explicit exclusion governance, materiality assessment, dependency mapping, anti-concealment rules, change control, exclusion propagation, and complete traceability, SEXM prevents unmodeled conditions from becoming invisible assumptions of coverage and prevents simulation results from expanding into conditions that were never simulated. By making absence itself governable, SEXM ensures that the limitations of a simulation remain as structurally visible as the conditions it successfully represents.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>